

(RELEASE 2)

C++ EQUALIZATOR PLUG-IN: USER'S MANUAL (ENG)

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1. FACILITY

This version (Release 2) is distributed in two formats:

- **VST3 Version (Plugin):** For use within DAWs (Ableton, Reaper, FL Studio, etc.).
Copy the .vst3 file to C:\Program Files\Common Files\VST3
- **Standalone version (Executable .exe):** Independent application with integrated audio player.

2. USER INTERFACE



The interface is divided into three main sections:

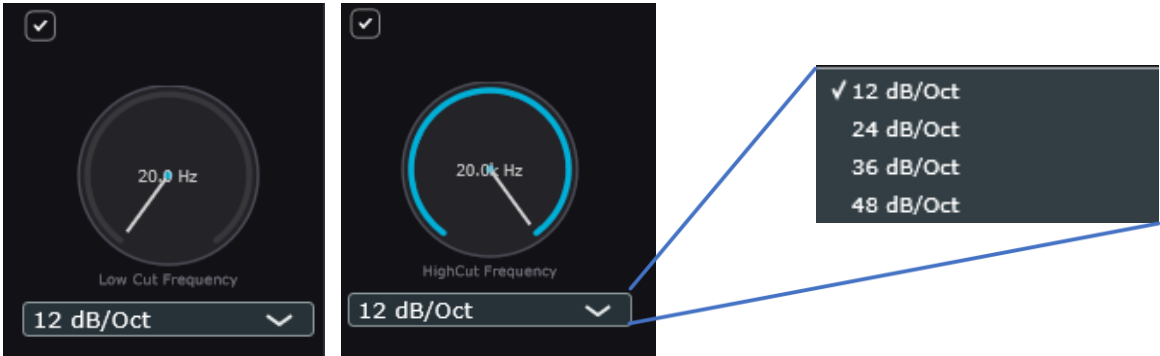
- **Controls Section (Lower):** Parameter settings by band.
- **Central Viewer:** Interactive spectrum and curve analyzer.
- **Utilities:** File player, default button, and appearance selector

3. EQUALIZER CONTROL

The plugin features 5 independent equalization bands:

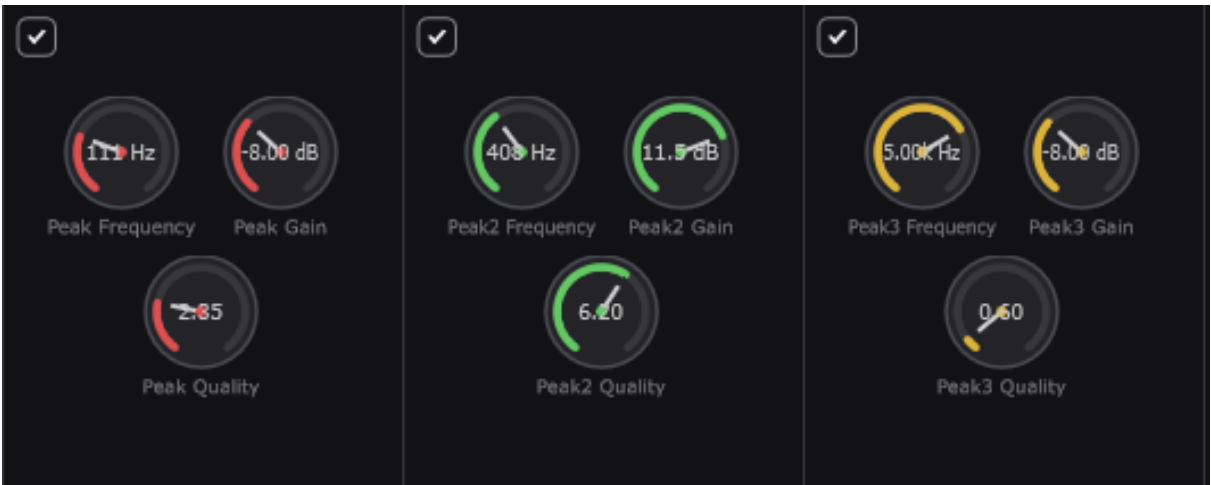
Cut-off Bands (High-Pass and Low-Pass Filters)

Control	Parameter	Description
Low Cut Frequency	Frequency (Hz)	Bass attenuation point.
High Cut Frequency	Frequency (Hz)	Treble attenuation point.
Slope	Slope (dB/Oct)	Slope selection: 12, 24, 36 or 48 dB/Oct.



Parametric Bands (Peak Filters)

Control	Parameter	Description
Peak Frequency	Frequency (Hz)	Selection of the center frequency.
Peak Gain	Gain (dB)	Boost or attenuation (± 24 dB).
Peak Quality	Q Factor	Bell bandwidth.



3.1. Graphical Parameter Editing

Starting with Release 2, all Peak band parameters can be adjusted intuitively directly on the response graph. This interactive method complements (and simplifies) the use of the lower controls:

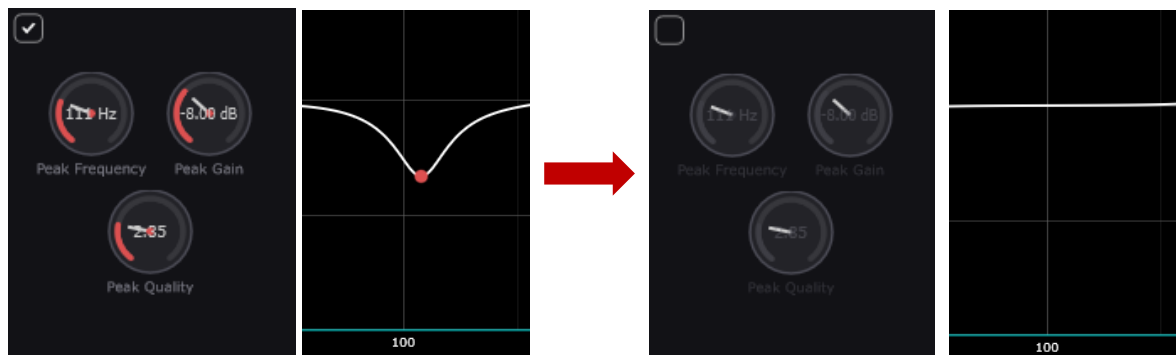


How to adjust the parameters with the mouse:

- **Detect a Peak:** Hover your cursor over any of the colored points on the graph (Red, Green, or Yellow). When you are close enough (approximately 30 pixels), the point will be activated and ready for editing.
- **Adjusting the Frequency:** Drag the dot horizontally (left-right) to change the filter's center frequency. The scale is logarithmic, meaning the control is consistent across the entire spectrum, from bass to treble.
- **Adjusting the Gain:** Drag the dot vertically (up-down) to increase or decrease the filter gain. Up = boost, down = cut.
- **Adjusting the Q-Factor (Bandwidth):** Use the mouse wheel while the pointer is active. Holding down the *Control* key while using the wheel triples the sensitivity for faster changes. Without *Control*, you'll get more precise adjustments.

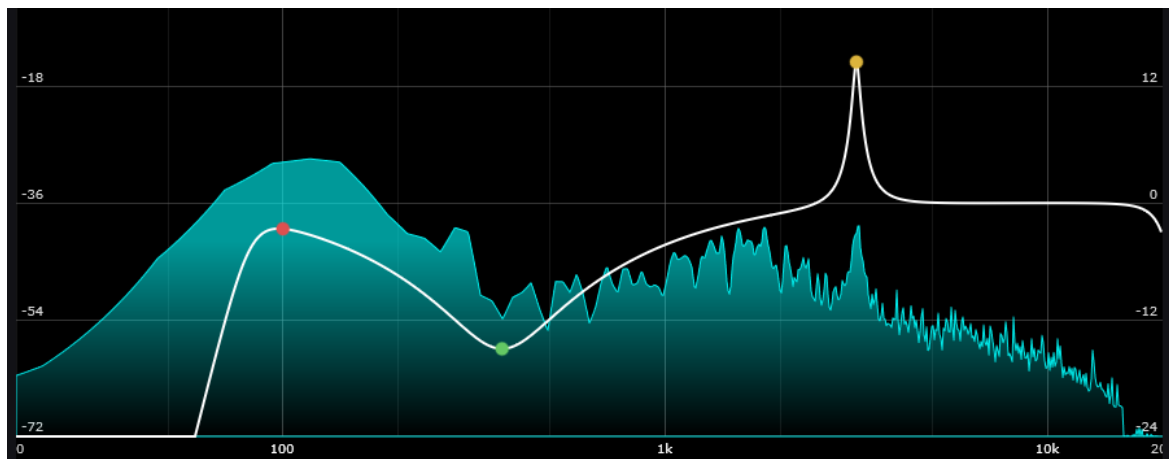
4. ACTIVATION SYSTEM (ON/OFF) AND BYPASS

Each of the 5 bands has an independent activation button located at the top of its control column.



- **On/Off Function:** These switches allow you to instantly activate or deactivate (Bypass) the processing of each filter.
- **Parameter Memory:** When a filter is deactivated, the plugin retains the potentiometer settings (frequency, gain, and Q). This allows for quick A/B comparisons to evaluate the impact of each band on the sound without losing the adjustments made.
- **Visual Status:** The equalizer uses an intelligent attenuation system. When a filter is turned off, the corresponding potentiometers and labels reduce their opacity and color. This visual feedback allows you to identify at a glance which bands are processing audio and which are at rest.

5. CURVE VIEWER AND SPECTRUM ANALYZER



Dynamic Response Curve

The viewer functions as a real-time response monitor. The colored nodes on the screen serve as visual guides that automatically move according to the values the user adjusts in the aforementioned controls.

- **White curve:** represents the total sum of all active filters.
- **Nodes:** indicate the exact position (frequency and gain) of each Peak filter, allowing immediate visual validation of the applied processing.

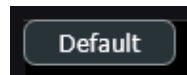
Spectrum Analyzer

It incorporates a 2048-point FFT (Fast Fourier Transform) that maps the energy of the background audio signal. This allows you to visually identify resonances or energy gaps to help you decide which parameters to adjust in the lower controls.

6. GLOBAL CONTROLS

Default button

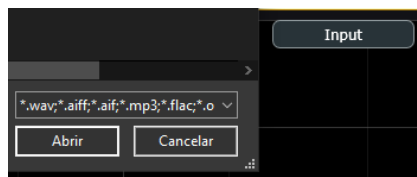
Resets all filters to their default position, in which no filtering is performed on the input



Audio playback controls

Standalone mode allows you to process external audio files without the need for a DAW:

- **Load Audio:** Opens a file explorer to load WAV, MP3, AIFF, etc. files.
- **Play Button:** Play/Stop controls for listening.
- **Gain Slider:** Adjusts the input level of the loaded track.



Appearance selector

Drop-down menu that allows the user to select their preferred appearance from 3 options:

